A Discussion About Popular Project Inputs, Outputs, Connectors, and Controllers

Prepared by Karen Craigs for Seneca College, © 2022-23 (v.1.2)

Audience: Seneca College electronics students, especially those taking TPJ452, TPJ653, TPJ655 or similar.

All photos in this document were taken either from DigiKey.ca or from the device's company website.

Note: Deciding which devices to use in your TPJ project will depend significantly upon the number and types of available GPIOs on your controller unit.

The following inputs, outputs, connectors, controllers, and software will be discussed in this document.

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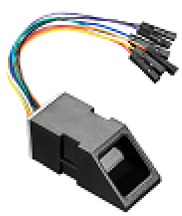
INPUTS

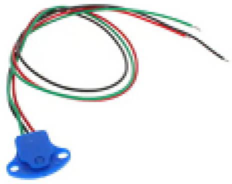
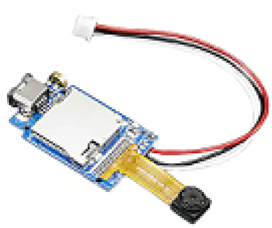
Often called sensors, since they receive data from the environmental and convert it into a signal. There are other types of sensors not listed here, but this should get you started thinking about your project.

Simple

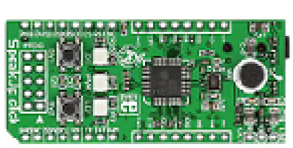
	Tactile or Limit Switch eg. SW401-ND (shown), P8062S-ND, discover many others PCB Footprint: space apart two **CAP-CC06**, or similar These momentary switches will likely need a pull-up or pull-down resistor to avoid a floating voltage value when not pressed. A limit switch is useful to gently determine physical limits or boundaries.
	Other Switches eg. CWI475-ND (shown), or others PCB Footprint: check the datasheet for dimensions Are you looking for toggles, slides, DIPs, rockers, or others? There are many sizes, shapes, colours, and basically they all trigger on or off.
	Touch Pad eg. 1738-SEN0297-ND (shown), or similar PCB Footprint: **LED**, or similar Press the pad to trigger their pressure sensitive switch.
	Conductive Thread eg. 1568-1804-ND (shown), or similar Not exactly an input but it is useful for sewing electronic wearables.
	Transistor eg. 2N3904 (NPN), 2N3906 (PNP), BS170 (FET), and many more! PCB Footprint: **SIP-3P**, **2N3904**, or similar Able to amplify a signal, or be used simply as an electronic switch. Useful for PWM control from digital IO pins that can't source current. BJTs use a resistor at the base, where FETs handle higher currents.
	Opto-Isolator eg. 160-1300-5-ND (shown), or similar PCB Footprint: **DIP6**, or similar These kinds of electronic switches can allow one circuit to trigger a second isolated circuit by using an internal LED and phototransistor.

Object Presence or Visual Sensing

	Photoresistor eg. 1528-2141-ND (shown) PCB Footprint: **SIP-2P**, or similar These optical sensors detect the presence (or absence) of light.
	Optical Sensor eg. 160-2210-ND (shown), or similar PCB Footprint: space apart two **SIP-2P**, or similar A transmitter is aimed towards a receiver between the U-shape unit. Place something opaque inside the U crevice to block the signal. Imagine holes around a round disc and counting each wheel step.
	Infrared (IR) Sensor eg. 1568-1272-ND (shown) PCB Footprint: **SIP-3P**, or similar These types of sensors can detect proximity to an object. An IRLED emits a particular wavelength of light to bounce off an object and a photodiode senses the same light. Many styles of units are available.
	Passive Infrared (PIR) Sensor eg. 1528-4871-ND (shown), 1528-1991-ND, or similar PCB Footprint: **SIP-3P**, or similar These types of sensors detect motion and often use a light diffuser.
	Oximeter or Heartrate Sensor eg. 2221-U029-ND (shown), 1738-SEN0344-ND, many others These types of sensors can measure your heartrate when placed next to skin, like by your thumb or inside wrist or clipped to your ear.
	Fingerprint Sensor eg. 1528-4690-ND (shown), or similar PCB Footprint: **SIP-8P**, or similar These types of sensors can scan your fingerprint. The software you write can be used to save several scans on file as part of a database.
	Ultrasonic Sensor eg. 1528-2832-ND (shown) PCB Footprint: **SIP-4P**, or similar These types of sensors measure the distance away from an object.

	Capacitive Displacement Sensor eg. 1835-PGN-SP-003-ND (shown), 1738-SEN0447-ND, or similar PCB Footprint: check the datasheet These change their capacitive field based on proximity to an object.
	Hall Effect Sensor eg. AH173-PL-A-ADICT-ND (shown), many others available PCB Footprint: **SIP-3P**, **2N3904**, or similar These types of sensors can trigger by the close presence of a magnet. There are four main types: unipolar, bipolar, omnipolar, and linear. Imagine using several to count ball magnets rolling down a tube.
	Time of Flight Camera eg. 3074-TR-EVO-15M-I2C-ND (shown), 3074-TR-EVO-MINI-I2C-ND These sensors use infrared light to determine depth information. They are popular in drone technology for depth-mapping areas. Terabee SAS is one of the manufacturers making these products.
	Camera Module eg. 1528-1799-ND (shown), or similar PCB Footprint: **SIP-3P**, or similar These record video or static images. Some can detect infrared images. Your software will process the captured video or static images, for example to read the face of a die or a human, or to “see” in the dark.

Audible Sensing

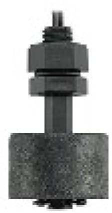
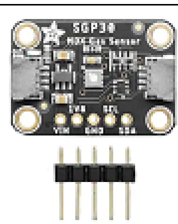
	Microphone eg. 102-6531-ND (shown), or similar PCB Footprint: **SIP-2P**, or similar These pick up audible noises and turn them into electrical signals. They must be placed with other parts (res, cap, etc.) in a specially designed circuit to work. They are often paired with an LM386 IC.
	Voice-Recognition Module eg. 1471-1334-ND (shown), 114991391-ND, many others Some modules come with on-board voice recognition capabilities, and will activate parts of your circuit with a unique voice command.

Mechanical Input

	Potentiometer eg. 3296W-102LF-ND (shown), or similar PCB Footprint: **SIP-3P**, **2N3904**, or similar Do you need to vary the voltage (pot) or vary the current (rheostat) in your design? Trimmer pots can typically have anywhere from 1 turn to 10 turns for their rotary knob, depending on the unit's precision.
	Tilt or Vibration Sensor eg. 1835-1000-ND (shown), 28036PAR-ND, or similar PCB Footprint: **SIP-2P**, or similar These sensors will trigger when tilted beyond a specified angle.
	Magnetic Reed Switch eg. 374-1092-ND (shown) PCB Footprint: **RES-1/4W**, or similar These switches will trigger when a magnet is placed nearby. Example: Imagine a magnet secured inside a housing unit with a reed switch in an external unit: the two units can come together and then trigger.
	Rotational or Shaft Encoder eg. PEC16-4215F-N0024-ND (shown), or similar PCB Footprint: modify 3/5 holes on a **SIP-5P**, **SIP-3P**, or similar These devices convert the position of a shaft into an electrical signal. They are also useful for selecting discrete items from a menu.
	Joystick eg. 27800-ND (shown), or similar PCB Footprint: space apart four **SIP-2P**, or similar These units often have two potentiometers tied to an x- and y-axis. Some models may include a pushbutton capability.
	Number Pad eg. 1528-1136-ND (shown), 27899PAR-ND, or similar PCB Footprint: **LED**, or similar 12- or 16-number keypads are commonly used for entry codes.
	Weight Scale Sensor eg. 1597-1482-ND (shown), or similar PCB Footprint: **SIP-4P**, or similar These types of force gauges detect and measure weight changes.

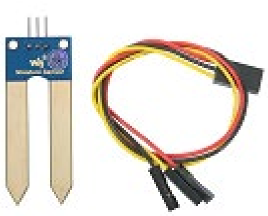
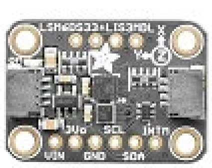
	Flex Sensor eg. 30056-ND (shown) PCB Footprint: **SIP-3P**, or similar These types of force convert flexibility into a resistance value.

Liquid- or Gas-Related Sensing

	Liquid Level Sensor eg. 1922-UM0034-000-ND (shown), or similar Some liquid level sensors use ultrasonic waves to visually determine the top of the liquid line. These can be top or bottom sensors.
	Float Level Sensor eg. 2010-1035-ND (shown), or similar Some liquid level sensors are activated with a float device (usually a foam piece) that lifts with the liquid to determine the top line.
	Liquid Turbidity Sensor eg. 235-1367-ND (shown) These types of sensors measure the relative clarity of a liquid.
	Flow Rate Sensor eg. 1597-1518-ND (shown), 2952-FS7.0.1L.195-ND, or similar These types of sensors measure the flow rate of the substance. There are liquid flow rate sensors and gas flow range sensors, for example.
	Gas Concentration Sensor eg. 1738-1093-ND (shown), many other types available PCB Footprint: **SIP-3P**, or similar These can sense multiple types of gasses, or can be quite specific.
	Air Quality Sensor eg. 1528-2531-ND (shown) PCB Footprint: **SIP-5P**, or similar Some models are multi-functional, measuring gas or humidity too.

	Pressure Sensor eg. MPX53GP-ND (shown), or similar PCB Footprint: **SIP-4P**, or similar These sensors measure the relative pressure via tube fixtures. There are liquid pressure sensors and gas pressure sensors (last is pictured).

Other Environmental Sensing

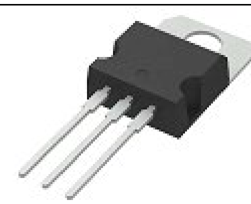
	Temperature Sensor eg. 296-43342-ND (shown), or similar PCB Footprint: **SIP-3P**, **2N3904**, or similar Different types of sensors can measure the ambient temperature in the area: thermistor, thermocouple, IR, resistance, semiconductor.
	Humidity Sensor eg. 235-1449-ND (shown), or similar PCB Footprint: **CAP-CC06**, or similar These types of sensors measure the ambient humidity in the area.
	Temperature & Humidity Sensor eg. 1597-101990561-ND (shown), or similar PCB Footprint: **SIP-3P**, **SIP-4P**, or similar You can often find inexpensive models that combine the two types.
	Barometric Pressure Sensor eg. 1738-1461-ND (shown) PCB Footprint: **SIP-4P**, **SIP-8P**, or similar These sensors measure the air pressure in the atmosphere.
	Moisture Sensor eg. 149-28092-ND (shown) , or similar PCB Footprint: **SIP-3P**, or similar These sensors are often used in soil to measure its moisture content.
	Accelerometer / Gyroscope / Magnetometer eg. 1528-4485-ND (shown), and many others available PCB Footprint: check the datasheet These sensors can permit motion, direction, and orientation sensing. Inexpensive versions are now available in 3-in-1 packages (shown).

	GPS Module You can use the standalone chips or the breakout boards for GPS, or "global positioning system". The latitude and longitude positional information is obtained from satellites orbiting the earth. These units need to use an antenna, a system that performs all the calculations and communications, and a device to display that information to you. Read: https://lastminuteengineers.com/neo6m-gps-arduino-tutorial/ Shown: 1597-109020022-ND, or 2221-U032-ND, 1738-TEL0137-ND

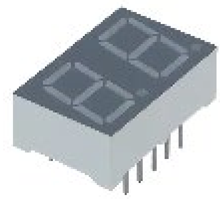
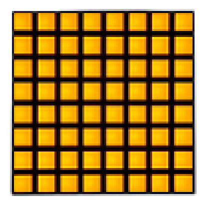
OUTPUTS

Sometimes called actuators, since the electrical signal may command a device to respond.

Simple

LED eg. C566D-RFE-CV0X0BB1-ND (shown), many others PCB Footprint: **LED**, **SIP-2P**, **DIODE**, or similar These light emitting diodes often need a series resistor to limit their current and prevent failure. Use them to indicate an ON or OFF state. Pictured is a red through-hole Radial package. Clear is also available.	
Voltage Regulator eg. 497-1443-5-ND (shown), many others PCB Footprint: modify 3/5 holes on a **SIP-5P**, **SIP-3P**, or similar These are useful for maintaining a fixed and stable DC voltage.	
DC-to-DC Converter eg. 2725-LSF01-K5B12SS-ND (shown), many others PCB Footprint: modify 4/5 holes on a **SIP-5P**, **SIP-3P**, or similar These are useful for converting a DC input voltage into an output DC voltage of higher or lower value.	
Transistor eg. 2N3904 (NPN), 2N3906 (PNP), BS170 (FET), and many more! PCB Footprint: **SIP-3P**, **2N3904**, or similar Able to amplify a signal, or be used simply as an electronic switch. Useful for PWM control from digital IO pins that can't source current. BJTs use a resistor at the base, where FETs handle higher currents.	

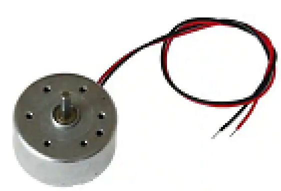
Visual Output

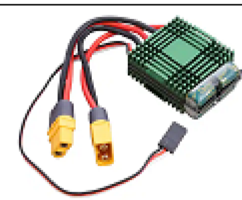
RGB LED eg. 1516-QBL8RGB25C0-ND (shown), or similar PCB Footprint: **SIP-4P**, or similar Check if the unit is common anode or common cathode. You will need to sink or source each of the three channels accordingly.	
LED Strip Lights eg. 1597-1672-ND (shown), reels, or similar PCB Footprint: **SIP-4P**, or similar These light strips are available in many lengths, colours, and setups.	
7-Segment Display eg. 160-1536-5-ND (shown), or similar PCB Footprint: space apart two **SIP-5P** or **SIP-9P** by 0.?in, or similar 16 LEDs are set within this popular 7-segment orientation used for numerals. A two-digit display may have 18 or 10 pins, for example. Multiplexing the digits may be required using software or an IC.	
8x8 LED Matrix eg. 1528-2227-ND (shown), or similar PCB Footprint: space apart two **SIP-8P** by 0.6in, or similar These outputs contain 64 individual LEDs as a matrix of 8+8 control pins, or reduce this down to 8+3 or even 3+3 pins using shift registers.	
OLED Display eg. 1597-104020208-ND (shown), or similar PCB Footprint: **SIP-4P**, **SIP-6P**, or similar These displays are quite versatile in their output, many using I2C/SPI. Some models have 4 or 6 pins, depending on the protocols used.	
LCD or TFT Display eg. 1528-2327-ND (shown), 1528-1345-ND, DFR0091-ND, or similar PCB Footprint: **SIP-16P**, or similar These displays are quite versatile in their output, many using I2C/SPI. Some popular models have 16 pins, but others have much fewer pins.	
Graphic Equalizer eg. 1568-1335-ND (shown), or similar PCB Footprint: **DIP8**, or similar (remember to use with an IC Socket) These types of chips simply divide the audio spectrum into 7 bands. Imagine turning sound signals into a colourful LED display.	

Audible Output

Buzzer eg. 445-2525-1-ND (shown), or similar PCB Footprint: **CAP-CC06**, or similar These devices make noise or “buzz” quite loudly, for example at 60db or louder. Transducers need an external PWM signal to drive it, where an indicator type is internally driven and accepts a DC voltage.	
Speaker eg. 433-1248-ND (shown), or similar PCB Footprint: **SIP-2P**, or similar These convert electrical audio signals into audible sounds waves. They must be placed with other parts (res, cap, etc.) in a specially designed circuit to work. They are often paired with an LM386 IC.	
Audio Player Module eg. 1568-1855-ND (shown), or similar PCB Footprint: check the datasheet Some audio playback devices can play WAV or MP3 type files.	

Mechanical Output

Relay or Relay Board eg. 2449-J104D2C5VDC.20S-ND (shown), or similar PCB Footprint: check the datasheet for layout These are switches, labelled as power (over 2A) or signal (under 2A). Discuss all connections to 120VAC devices with the professor first.	
AC or DC Fan eg. 1528-1904-ND (shown), or similar PCB Footprint: **SIP-2P**, **LED**, or similar These come in many shapes, sizes, and voltages but typically they are two- or three-pin devices that run with an AC or DC applied voltage.	
AC or DC Motor eg. 1927-1023-ND (shown), or similar PCB Footprint: **SIP-2P**, or similar These come in many shapes, sizes, and voltages. Some typical styles include vibration motors, geared motors, and standard motors.	

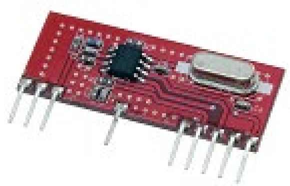
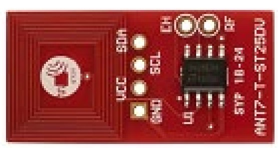
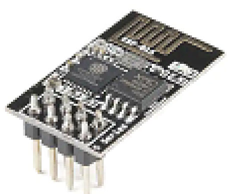
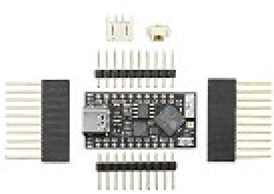
Servo Motor eg. 1738-1385-ND (shown), or similar PCB Footprint: **SIP-3P**, or similar These do come in different sizes and voltages but typically they are three-pin 5V devices that run with a particular PWM signal.	
Stepper Motor eg. 1528-1366-ND (shown), or similar PCB Footprint: **SIP-5P**, or similar These come in many shapes, sizes, setups, and voltages. These often require a motor driver board to run the correct order of signals.	
Driver Boards eg. 1597-1114-ND (shown), 1528-1795-ND, or similar PCB Footprint: check the types of connectors needed These boards help run large current devices (motors, lamps, pumps).	
Electronic Speed Controller (ESC) eg. 1738-1444-ND (shown) PCB Footprint: check the types of connectors needed These boards control DC motor speed, often used for remote control.	

Liquid- or Gas-Related Outputs

Liquid Pump eg. 1528-4547-ND (shown) PCB Footprint: **SIP-2P**, or similar These use a submersible pump to force liquid through a tube or pipe.	
Liquid Solenoid Valve eg. 1528-2003-ND (shown) PCB Footprint: **SIP-2P**, or similar These act like an electronic tap to open or close the valve in a pipe. There are lots of types, including miniature 3V, 5V, 6V (etc.) models.	
Vacuum or Air Pump eg. 1528-4699-ND (shown) PCB Footprint: **SIP-2P**, or similar These are attached to a DC motor and can suck or blow air into tubes.	

Pressure Valve eg. 1738-DFR0866-ND (shown) PCB Footprint: **SIP-2P**, or similar These are attached to a solenoid and act like a 3-way air valve.	

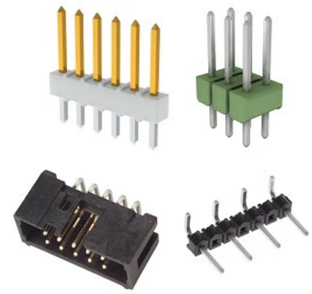
Other Types of Connectivity

	RF Transmitter/Receiver Modules Find a matching pair of modulation types (eg. 433.92MHZ, Rx shown) and use these paired devices to wirelessly communicate between nearby assemblies. Circuit designs are often inside the datasheets. Shown: AM-RX12E-433P-ND, eg. 1597-1223-ND, find other Rx and Tx
	NFC or RFID Paired Modules Find a matching pair of tag and reader (eg. 13.56MHz, reader shown) and use these paired devices to identify and limit user access into buildings or boxes. Circuit designs are often inside the datasheets. Shown: 497-18394-ND, find others
	Wi-Fi Module The ESP8266 chips are popular in Wi-Fi connectivity projects. There are lots of models with different features included at different price points, and they come in many sizes for all kinds of applications. Shown: 1568-WRL-17146-ND
	Bluetooth Module The ESP32 chip is the successor of ESP8266 and includes Bluetooth capability, more GPIOs, more sensors, and other improvements. For example, the HC-05 is a popular board that just uses the Bluetooth SPP wireless serial connection. Note in this picture: a tiny ESP32 board is soldered to the bigger board – when researching, figure out which one you need before ordering it for your design. Shown: 1597-1164-ND
	All-in-One MCU with Wi-Fi and Bluetooth Integration The low-cost, low-power microcontrollers of ESP32 and ESP8266, also referred to as a "system on chip" or SoC design, have a built-in MCU. This makes them perfect for IoT applications, and there are dozens of models to review. Imagine using tiny versions (eg. M5Stamp Pico) as interactive game pieces, or larger versions for remote connectivity. Shown: 1528-5028-ND, also 2221-K051-B-ND and 2221-C050-B-ND

	Real-Time Clock (RTC) Module These modules "give your project the ability to tell time, store time, or make time-based decisions with 1 second resolution in a small, compact form-factor." (from the datasheet) Shown: 29125-ND
	Breakout Boards These come in countless sizes, shapes, and purposes! The specific board(s) you have in mind for your project should be provided in your Team Proposal as it may affect the approval of your project. Shown: 1568-1721-ND
	Battery Management Boards The types of batteries you use in your project may need special management, for example for Li-ion (eg. 18650) and Li-Po batteries. Pay attention! Charging rates may affect the lifetime of the batteries, and some batteries can cause dangerous fires if they are mishandled. Shown: 1597-106990290-ND

CONNECTORS & CABLES

Proper cabling is expected inside and around your TPJ project. Choose the appropriate size and length and make your own connector cable assemblies if necessary. Some tools may be provided in the lab for this purpose, though teams are expected to purchase the cables and connectors themselves.

### Jumper Wires Do not use prototyping jumper wires in your final product assembly. These are suitable for prototyping your project at the breadboard stage but should not be present when you switch over to the PCB build about halfway through the semester! If you've designed your PCB appropriately, there shouldn't be any soldered wires on it.	
### Male Headers These are perfectly suited for your final project build and come in many sizes, shapes, and colours. "Header" refers to either the male or female style of this connector. You may wish to coordinate your PCB design such that different input and output connectors attach to different colour headers. Common headers have 2.54mm pitch. Recommended PCB Footprint: **SIP-6P**, **CON-SIP-6P**, or similar	
### Female Headers Don't use this option if you are planning to put jumper wires in them. Some have JST friction locks. Check pin pitch too, typically 2.54mm. These can replace IC Sockets if you can't source the right row spacing. Recommended PCB Footprint: **SIP-6P**, **CON-SIP-6P**, or similar	
### Ribbon Cable & Press-Fit Connectors These are perfectly suited for your final project build since you can choose the connector and cut the length of cable to suit your project. Pay attention to the pitch of the cable (shown 1.27mm) as well as the number of wires or pins. Shown are double-row IDC connectors, used with an IDC press-fit crimping tool. Other types: Molex, JST, Dupont. Shown: MD06R-1-ND, 1175-1658-ND, 732-5443-ND	
### IC Sockets These may be required on your PCB, for example if you are using an IC or a breakout board that already has header pins installed. Pay attention to the pin pitch, which is typically 2.54mm (0.1") or 5.08mm (0.2"), and row spacing, which is typically in increments of 0.1". Again, don't use with regular jumper wires on your final project. Shown: A120347-ND, ED90146-ND	

Screw Terminals and Wire-to-Board Connectors You may choose to install these or they may be installed on the controller you are using. Pay attention to the pitch, which is typically 2.54mm or 5.08mm. Don’t use just with jumper wires: install ferrules, headers, or another type of connector to insert into the terminal. Shown: ED2611-ND, 277-13752-ND	
Ferrules Like the bit of plastic at the end of a shoelace, a ferrule is used to gather up the end of a stranded wire into a tight metal sleeve. Ferrules require crimpers to install. Pay attention to their gauge. Shown: 288-1046-ND	
Heat Shrink Tubing These are great for use in your final product assembly. They provide wire management and carefully cover over soldered connections. They can add an element of professionalism to your final PCB build. Shown: A124391-DS-ND	
Zip-Ties and Other Cable Management These are great for use in your final product assembly. They provide wire stability and security in your project. They can add an element of professionalism to your final PCB build. Shown: 145-22CC37C0125B-ND, RPC5678-ND	

CONTROLLERS

Choose a controller for your TPJ project. Note the number and types of ports supported for your project. Often the controller will need a separate power supply if you have current-demanding output devices. **Make good connectors**. Don't use single wires to each GPIO port. Instead, bundle several nearby wires going to your unique PCB. Seneca Wifi might work with a 5 GHz bandwidth MCU, depending on settings.

Suggestions are listed here for your ideation process. Separate documents review many more boards.

SIMATIC IOT2050

The IOT2050 could make an excellent choice for TPJ653 projects. These devices are already installed in Lab K2200 and offer 4 analog inputs, 4 digital inputs, and 2 digital outputs for peripheral devices. They easily connect PLCs to the cloud with a UI such as Node-RED.

You may test your parts using the IOT2000 board installed in the lab. Your final prototype will require good connectors going to your PCB.

FRDM K64

The FRDM K64 could make an excellent choice for TPJ452 projects. These devices are already installed in Lab A4072 and can offer 6 analog ports and 14 digital ports (eg. up to 5x UARTs, 2x SPIs, 2x I2Cs and 1x CAN connected to Headers as multiplexed peripherals). Try using Node-RED as UI to the cloud or write your own HTML platform.

You may test all your parts using the Base Shield installed in the lab. Your final prototype will **only** connect to the shield's female headers, not the JST type connectors. i.e. Make good connectors to your PCB.

ESP32-S3 DEVKITC

The ESP32-S3 could make a good choice for any portable TPJ project. These devices are inexpensive, easy to source, and easy to learn with online Getting Started documentation. They offer 34 GPIOs including 4 SPI, 3 UART, 2 I2C, 2 I2S, and 12-bit ADC for peripheral devices. They are easy to code through Visual Studio & PlatformIO software. Search on DigiKey.ca for "esp32-s3-devkitc" to find several versions.

Raspberry Pi

In the past, Raspberry Pi's were good choices for use in TPJ projects. However, they are costly, easy to kill, and difficult to source now. Note that the Zero and Pico models will likely not have enough ports to interface with all your peripheral devices, and not all models have Wifi or Bluetooth. RPi 3 tends to overheat and brick.

Therefore, RPi models are not recommended for use this semester.

BeagleBoard

A BeagleBone board may be a suitable choice for your TPJ project, depending on the model chosen. However, they have the same cautions as the Raspberry Pi models. The BeagleBone Blue is a robotics controller whereas the BeagleBone Black Wireless has built-in wireless networking capability. Find out all its GPIO types first.

When looking for these and (eg. 30+) other similar boards, search for "Single Board Computers" as the board type. You can also search under the "Evaluation Boards – Embedded – MCU, DSP" page.

Arduino

Avoid using Arduino boards in your project assembly.

These include: UNO, DUE, MEGA, Leonardo, or any other brands with similar architecture or connectivity to the Arduino IDE.

DigiKey.ca lists these board types under "Evaluation Platform".

Application Board Kits

Avoid using board kits as these are often unsuitable for TPJ projects. The point of TPJ is to make the connections to the peripheral devices yourself, with a PCB design by your team. Some exceptions can be made during the proposal process, depending on the board's usage and other devices being used in the project.

Remote Controllers

There are many on the market! A popular model that is reliable and easy to source now because of the booming drone industry is the FlySky 6-channel RC system, which uses the FS-i6X transmitter and FS-iA6B receiver. Installing an RC controller in your TPJ project would be considered as a secondary controller, not primary.

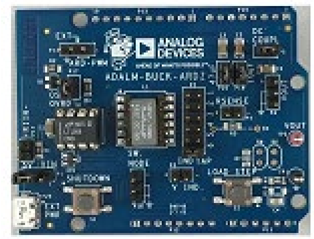
While it is fun to use manual remote control, consider that it also does not directly address the issue of automation in your project.

SHIELDS, HATS, GROVE BOARDS, ETC.

These pre-made boards typically have built-in interfacing capabilities with other popular devices.

Shields

These should not be used in your product assembly. Since you are creating your own PCB design, you are in effect making your own breakout board and would have no use for a pre-made board.
An exception can be made in your Project Contract with permission, but any parts used on it would not count for any hardware IO.



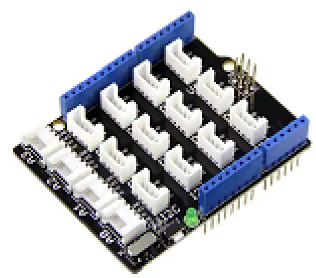
Hats

These should not be used in your product assembly. Since you are creating your own PCB design, you are in effect making your own breakout board and would have no use for a pre-made board.
An exception can be made in your Project Contract with permission, but any parts used on it would not count for any hardware IO.

Grove Boards

These should not be used as a primary board in your project. Since you are creating your own PCB design, you are in effect making your own breakout board.

If you are using the FRDM K64 MCU board, this Grove Shield is fixed in place in the lab. Consider how to design your PCB around them.



LCD Display Touch Screens

These may be used with your project, but they come with a caution. These units typically take up all the pins to your controller, even though they don't <u>use</u> all the pins. If you set up a breakout cable from your MCU to a PCB first, which then provides a separate socket to the LCD display, you might be able to use it – at the very least, it must be powered separately. You would therefore need to play around with this setup to check if it is suitable for your project. It will also likely require a separate power supply from the rest of your project.

SOFTWARE AND CODING

TPJ requires you to write program code to interface inputs and outputs with the controller and an **IoT** (Internet of Things) platform. Your project will need to connect to the internet in some way.

Use **C/C++** programming for your TPJ Project, since you already have had training with it (not Python). Any external coded libraries used to interface hardware devices should be approved first before using them – ask the instructor. (eg. Can't use "Arduino.h" or similar.) **Write your own functions, cite all resources** used to develop your project code, and **provide fully annotated comments.** More hints:

Visual Studio This suite is perfectly suited to develop your TPJ project **C/C++ code**. It is free to download at home and is also available through MyApps. You can then install the PlatformIO extension, which will interface with your GPIO ports on your MCU. Look for tutorials of PlatformIO with C/C++ code, eg: AbIoTs Lab, mothunderz, tecteach2489.	
TIA Portal This suite is perfectly suited to develop your TPJ project **PLC code**. It is available to use through MyApps.	
Mbed Online IDE This suite is perfectly suited to develop your TPJ Project C/C++ application on an Mbed OS (eg. FRDM K64, NXP MCU, etc.). Note that you'll likely use the **Keil Studio Cloud** instead of MCUExpresso, since the latter no longer has access to new program creation (2023).	arm MBED
Node-RED This is a free web browser-based development tool for visual programming. It comes with a tool for wiring the IO devices and can generate its own IoT dashboard user interface. Be sure to pick the right elements for device control as well as monitoring inputs.	
IoT Dashboard? There are other platforms that can connect your project to the internet and display and control your project. Unfortunately, they don't all provide internet-based device control, which is required. Note: Blynk, ThingsBoard, and similar will not be approved for use.	
Arduino IDE Since you are **not using an Arduino** for your main project assembly, it would be unsuitable to use the Arduino IDE to create your main code. However, some exceptions may be approved in your TPJ proposal, for example for secondary boards running remote applications. **Avoid using "Arduino.h"** -- eg. define your own board info, etc.	